WORKSHEET - NAPLAN Year 9

Statistics and Probability
- **Data and Statistics**
- **Probability**

Geometry
- **Maps and Bearings**
- **Ratio and Scale in Geometry**
- **Triangles and Other Geometric Properties**

Teacher: Brad Summers

Exam Equivalent Time: 40 minutes *(based on NAPLAN allocation of ~1.25 min/mark)*

Questions

1. Geometry, NAP-08569

The distance between Newcastle and Canberra on the map below is 9 cm.

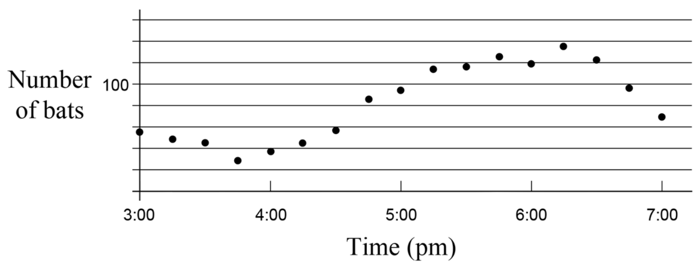
The scale of the map is 1 cm = 50 km.

What is the actual distance between Newcastle and Canberra?

350 km   450 km   600 km   900 km

2. Statistics and Probability, NAP-19140

This graph shows the number of bats in a fruit tree at 15 minute intervals over 4 hours.



At which time were the highest number of bats in the fruit tree?

5:45   6:00   6:15   6:30

3. Statistics and Probability, NAP-32090

Arun flips an unbiased coin 200 times.

Which result is most likely?

20 tails   98 tails   108 tails   196 tails

4. Statistics and Probability, NAP-25056

The probability of rolling a 6 on a standard six-sided die is

$\frac{2}{3}$ $\frac{1}{2}$ $\frac{1}{6}$ $\frac{1}{5}$

5. Statistics and Probability, NAP-55612

10 of the tallest mountains in the United States are listed in the table below:

Name of Mountain	Height (m)	Location
Denali	6190	Alaska
Mount Bona	5044	Alaska
Mount Massive	4398	Colorado
Castle Peak	4352	Colorado
Mount Evans	4350	Colorado
Mount Gabb	4190	California
Mauna Loa	4169	Hawaii
Kings Peak	4125	Utah
Wheeler Peak	4013	New Mexico
Mount Bear	1540	Alaska

How much taller than Utah's tallest mountain is Colorado's tallest mountain?

225 metres 227 metres 273 metres 330 metres

6. Statistics and Probability, NAP-49697

Claudia gets to ring the school bell once every 5 school days.

Today is a school day.

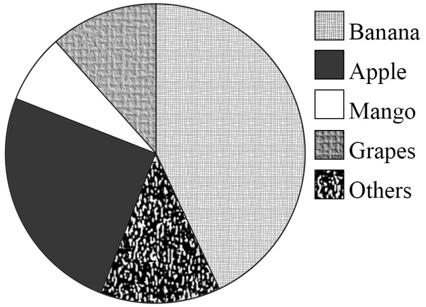
What is the probability that Claudia will ring the school bell?

5% 0.35 $\frac{1}{5}$ $\frac{5}{7}$

7. Statistics and Probability, NAP-08570

Anne taught a kindergarten class and asked her students what their favorite fruit is.

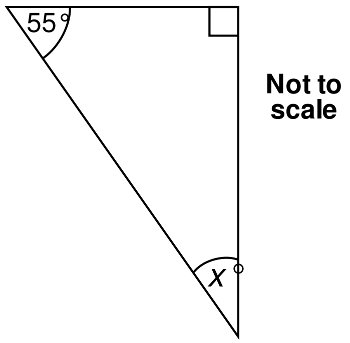
Their answers were used to draw the pie chart below.



Around a quarter of her students preferred which fruit?

Banana Apple Mango Grapes

8. Geometry, NAP-25057



What is the value of x?

35 45 55 65

9. Statistics and Probability, NAP-48579

Ricky's last 10 cricket scores are recorded below.

3, 15, 21, 21, 28, 30, 31, 38, 38, 40

He put these scores into a stem-and-leaf plot.

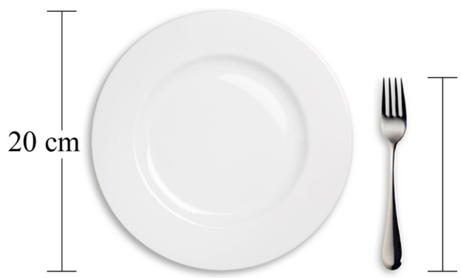
Stem	Leaf
4	0
3	1 8 8
2	1 1 8
1	5
0	3

KEY
4|0 = 40 runs

Which score is missing from the plot?

10. Geometry, NAP-25086

Danny was eating dinner and after finishing he took a photo of his plate and fork.



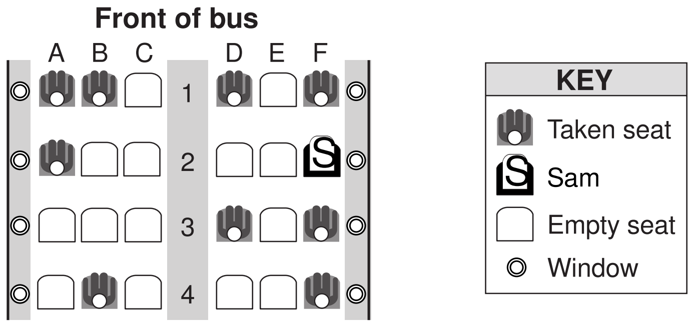
Which of the following measurements is closest to the length of the fork?

9 cm 10 cm 12 cm 15 cm

11. Geometry, NAP-95623

Here is a seating plan for a tourist bus.

Sam is seating in the window seat 2F.



Chilla wants to book a window seat as close as possible to the front of the bus.

Which empty seat should Chilla book?

1C 2B 3A 4A

12. Statistics and Probability, NAP-49696

Five students do a standing long jump at their athletics carnival and the length of their jumps, in centimetres, are recorded in the table below.

Student	Rob	Mary	Stew	Lenny	Kath
Distance (in cm)	70	75	66	95	50

If Kath's distance is removed from the data, what happens to the mean distance that is jumped from this group?

- It increases
- It decreases
- There is no change
- It is not possible to tell

13. Statistics and Probability, NAP-13225

A bag contains 30 marbles that are either blue or green.

The probability of randomly removing a blue marble is 30%.

What is the probability of randomly removing a green marble?

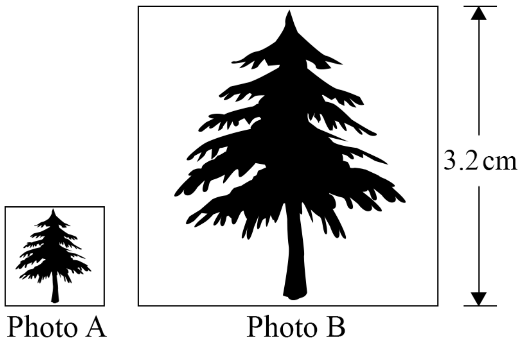
[ ] %

14. Geometry, NAP-55616

Juanita wanted to print a larger copy of her photo of a tree.

She increased the scale by a factor of 3.

She named the first picture Photo A and the second picture Photo B.



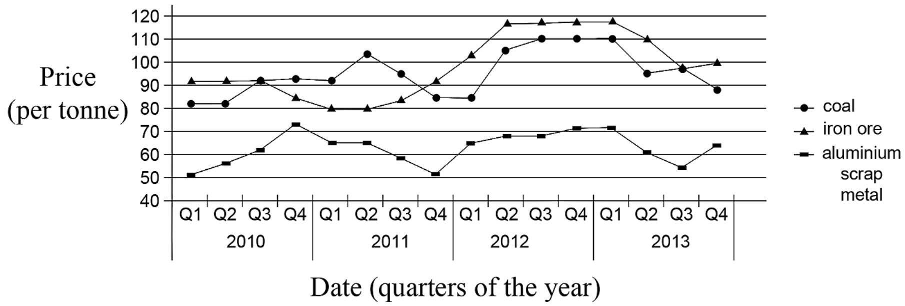
If Photo B is 3.2 cm high, how high is Photo A?

0.8 cm ○ 1.12 cm ○ 1.07 cm ○ 9.6 cm ○

15. Statistics and Probability, NAP-36751

This graph shows the price per tonne of resources in a country.

Resource prices



The price of coal was greater than the price of iron ore for a number of quarters, as shown by the graph.

In one of those quarters, the price of aluminium scrap metal fell below $60 per tonne.

Which quarter was this?

2011 Q1 ○ 2011 Q2 ○ 2011 Q3 ○ 2011 Q4 ○

16. Statistics and Probability, NAP-08599

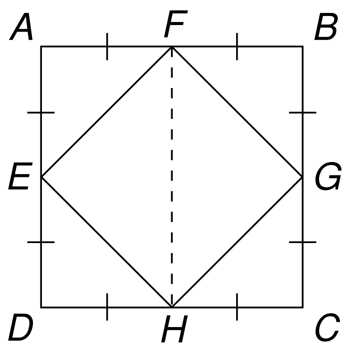
The weatherman described the chance of thunderstorms on a given day as highly unlikely.

Which probability best describes the chance of a thunderstorm?

$\frac{1}{11}$ ○ $\frac{1}{2}$ ○ $\frac{10}{11}$ ○ 1 ○

17. Geometry, NAP-66189

Two squares $ABCD$ and $EFGH$ are shown below.



The area of triangle FGH is

- quarter the area of $ABCD$
- four times the area of $ABCD$
- one eighth the area of $ABCD$
- eight times the area of $ABCD$

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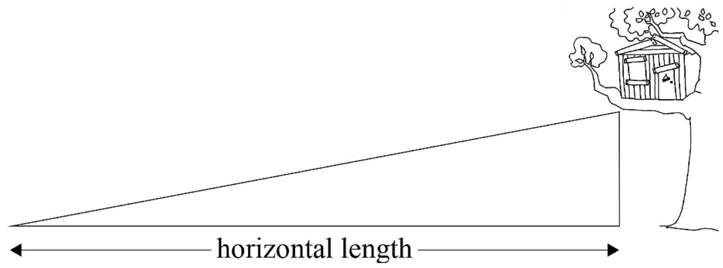
18. Geometry, NAP-07313

Vinny made a ramp to access his tree house that is 1.5 metres high.

The ratio of the horizontal length of the ramp to its height is 8:1.



What is the horizontal length of the ramp?

8 metres | 12 metres | 16 metres | 24 metres

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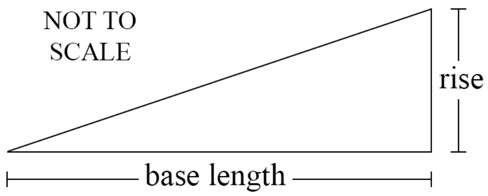
19. Geometry, NAP-32121

A ramp rises 1 metre for every 8 metres of its base length.



If it rises 0.7 metres, what is its base length?

5.6 metres | 8.7 metres | 11.4 metres | 16 metres

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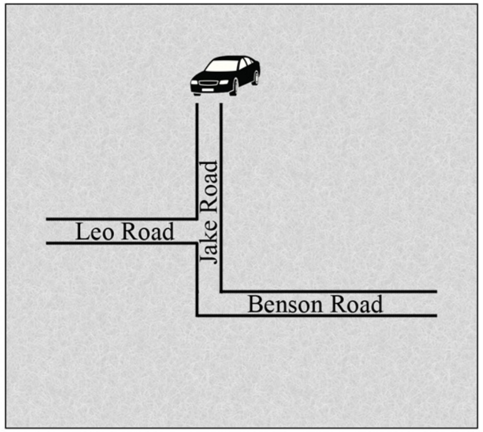
20. Geometry, NAP-01564

A car travelling south-west along Jake Road takes a 90° turn left into Benson Road.



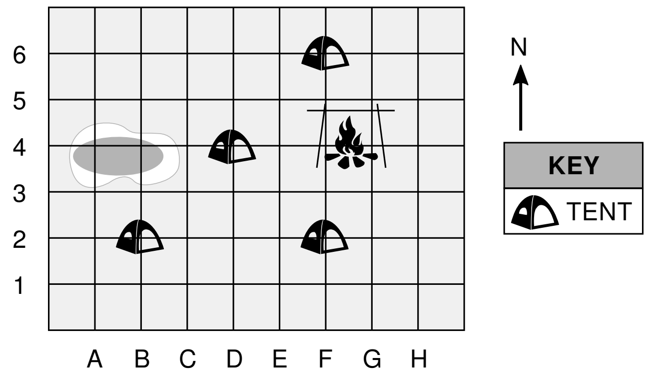
In which direction will the car be travelling along Benson Road?

- south-west
- north-east
- south-east
- north-west

21. Geometry, NAP-26205

Cameron is camping with 4 friends.

They have individual tents shown on the diagram below.



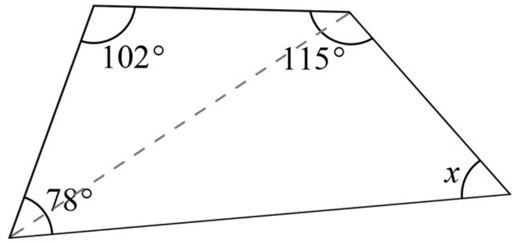
Brandon's tent is north-west of Cameron and Dan's tent is to Cameron's north.

What is Cameron's position?

- B2
- B6
- D4
- F2
- F6

22. Geometry, NAP-73250

What is the size of the angle marked x in the diagram below?

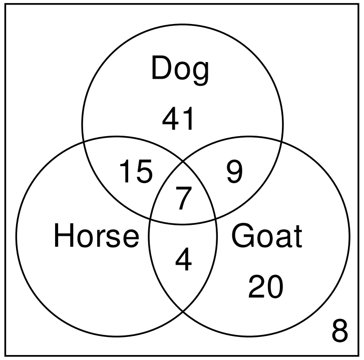


degrees

23. Statistics and Probability, NAP-90859

A country school surveyed 120 of its students about the type of animals they have at home.

The results are recorded in the Venn diagram below, although the number of students who only own horses is missing.



If one of the students is selected at random, what is the probability that the student does not own a goat?

$\frac{40}{120}$ $\frac{60}{120}$ $\frac{80}{120}$ $\frac{100}{120}$

24. Number, NAP-26208

The dog population in five continents can be found in the table below.

Continent	Dog population
Africa	9, 721, 457
Asia/Oceania	72, 428, 799
Europe	59, 329, 647
North America	43, 520, 201
South America	13, 262, 582

How many **more** dogs live in Europe than in Africa and South America combined?

25. Statistics and Probability, NAP-95650

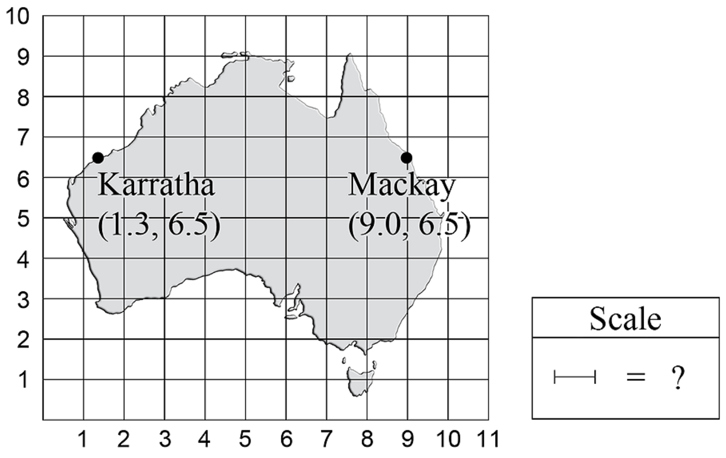
Shane rolls a standard six-sided dice.

Which of the following events has a probability of less than 0.5?

- rolling a number greater than 1
- rolling an odd number
- rolling a number less than 6
- rolling a number greater than or equal to 6

26. ☒ Geometry, NAP-60300

Alison draws a scale map of Australia on a coordinate grid.



She knows that the actual distance between Karratha and Mackay is 2950 kilometres.

Which scale value should Alison use for her map?

180 km ⬭ 330 km ⬭ 400 km ⬭ 500 km ⬭

27. ☒ Statistics and Probability, NAP-07344

The mean height of students at a Year 9 class is 155 cm.

The median height of the same class is 145 cm.

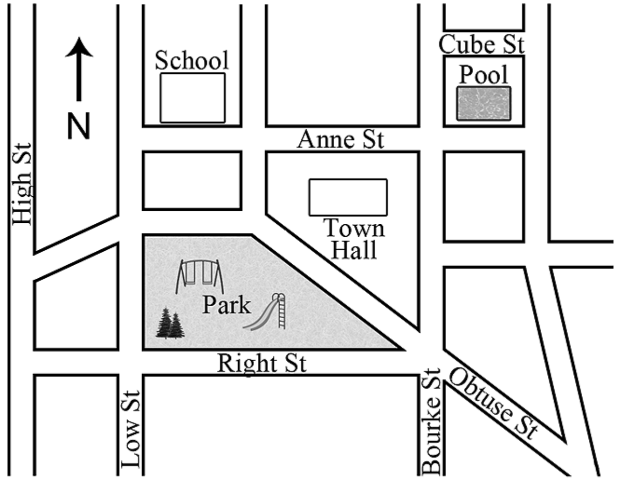
A new student joins the class who is 165 cm tall.

Which of these is possible?

- ⬭ The mean height stays the same and the median height increases.
- ⬭ The mean height stays the same and the median height decreases.
- ⬭ The mean height decreases and the median height increases.
- ⬭ The mean height increases and the median height stays the same.

28. ☒ Geometry, NAP-89740

Harry lives on a street that runs directly east-west.



Harry's house is east of town hall and north of the pool.

What street does Harry live in?

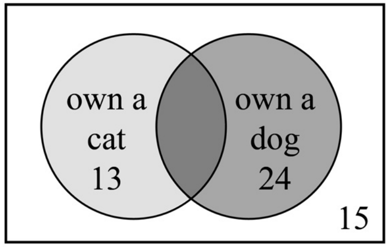
Right St. ⬭ Bourke St. ⬭ Anne St. ⬭ Cube St. ⬭

29. Statistics and Probability, NAP-49759

Sixty Year 9 students were surveyed about their pets.

- 32 owned a dog
- 21 owned a cat
- 15 didn't have a dog or a cat

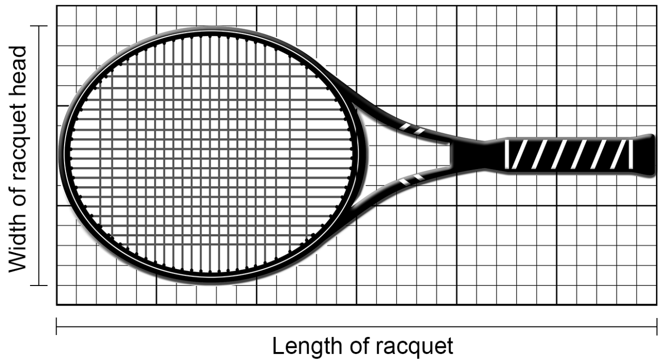
The diagram is missing some information.



How many students own both a dog and a cat?

30. Geometry, NAP-32151

Jana drew a diagram of her tennis racquet.



The actual length of her racquet is 60 centimetres.

What is the actual width of her racquet head?

centimetres

31. Statistics and Probability, NAP-19200

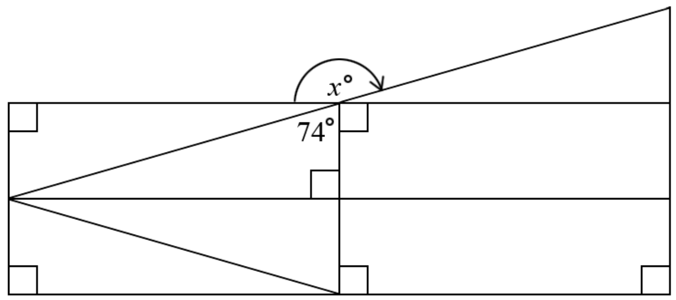
A standard six-sided die is rolled once.

What is the probability that the number on the top face is a factor of 4?

$\frac{5}{6}$ $\frac{1}{2}$ $\frac{1}{3}$ $\frac{1}{6}$

32. Geometry, NAP-61590

Brett drew the shape below using rectangles and straight lines.



What is the value of x?

degrees

Worked Solutions

1. Geometry, NAP-08569

$$\begin{aligned}\text{Distance} &= 9 \times 50 \\ &= 450 \text{ km}\end{aligned}$$

2. Statistics and Probability, NAP-19140

The highest data point is one interval past 6:00 pm.

$\therefore$ The highest number were in the tree at 6:15 pm.

3. Statistics and Probability, NAP-32090

98 tails

4. Statistics and Probability, NAP-25056

$$\begin{aligned}P &= \frac{\text{favorable events}}{\text{total possible events}} \\ &= \frac{1}{6}\end{aligned}$$

5. Statistics and Probability, NAP-55612

$$\begin{aligned}&\text{Mount Massive} - \text{Kings Peak} \\ &\quad = 4398 - 4125 \\ &\quad = 273 \text{ metres}\end{aligned}$$

6. Statistics and Probability, NAP-49697

$$\begin{aligned}P &= \frac{\text{favorable events}}{\text{total possible events}} \\ &= \frac{1}{5}\end{aligned}$$

Worked Solutions

7. Statistics and Probability, NAP-08570

A quarter of students will represent a sector that is the size of a quarter of the circle.

$\therefore$ Apple

8. Geometry, NAP-25057

$$x = 180 - (55 + 90)$$
$$= 35$$

9. Statistics and Probability, NAP-48579

30 is missing.

10. Geometry, NAP-25086

The fork is approximately 75% the height of the plate,

$$\therefore \text{ Fork height} = 75\% \times 20$$
$$= 15 \text{ cm}$$

11. Geometry, NAP-95623

$3A$

12. Statistics and Probability, NAP-49696

The mean increases because the shortest distance is removed from the data set.

13. Statistics and Probability, NAP-13225

$$P(G) = 1 - P(B)$$
$$= 1 - 0.3$$
$$= 0.7$$
$$= 70\%$$

14. Geometry, NAP-55616

Since Photo B is increased by a factor of 3,

$$\frac{\text{Height of Photo B}}{\text{Height of Photo A}} = 3$$

$$\therefore \text{ Height of Photo A} = \frac{\text{Height of Photo B}}{3}$$
$$= \frac{3.2}{3}$$
$$= 1.07 \text{ cm}$$

15. Statistics and Probability, NAP-36751

2011 Q3

16. Statistics and Probability, NAP-08599

$$\frac{1}{11}$$

17. Geometry, NAP-66189

$$\text{Area } EFGH = \frac{1}{2} \text{ Area } ABCD$$

$$\text{Area } \Delta FGH = \frac{1}{2} \text{ Area } EFGH$$

$$\therefore \text{ Area } \Delta FGH = \frac{1}{4} \text{ Area } ABCD$$

18. Geometry, NAP-07313

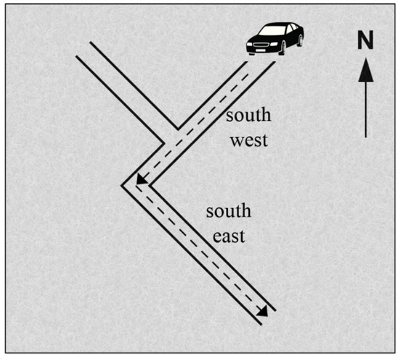
$$\text{Ratio} = 8:1$$
$$= (8 \times 1.5):(1 \times 1.5)$$
$$= 12:1.5$$

$\therefore$ Horizontal length is 12 metres.

19. Geometry, NAP-32121

$$\text{Base length} = 8 \times 0.7$$
$$= 5.6 \text{ metres}$$

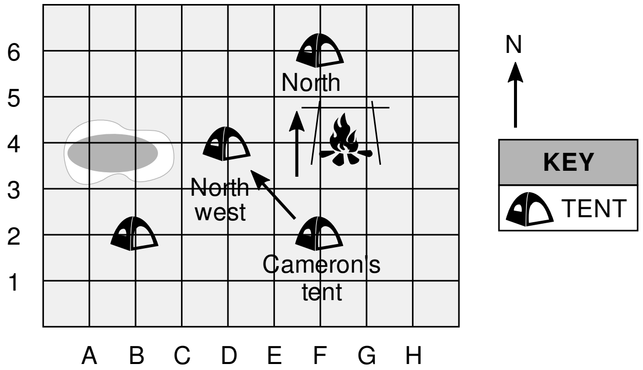
20. Geometry, NAP-01564



south-east

21. Geometry, NAP-26205

Cameron's tent is located at F2.



22. Geometry, NAP-73250

$$\text{Sum of internal angles} = 360^\circ$$
$$\therefore x = 360 - (102 + 115 + 78)$$
$$= 65^\circ$$

23. Statistics and Probability, NAP-90859

Number of students who do not own a goat

$$= 120 - (9 + 7 + 4 + 20)$$
$$= 120 - 40$$
$$= 80$$
$$\therefore \text{ Probability} = \frac{80}{120}$$

24. Number, NAP-26208

Number of extra dogs

$$= 59\,329\,647 - (9\,721\,457 + 13\,262\,582)$$
$$= 36\,345\,608$$

25. Statistics and Probability, NAP-95650

$$P(\text{number} \geq 6) = \frac{1}{6},$$

which is less than 0.5.

26. Geometry, NAP-60300

Mackay is 7.7 grid intervals away from Karratha.

$$\therefore \text{ Scale value} = \frac{2950}{7.7}$$
$$\approx 400$$

27. Statistics and Probability, NAP-07344

It is possible for the mean height to increase and the median height to stay the same.

28. Geometry, NAP-89740

Cube St

29. Statistics and Probability, NAP-49759

Since 32 students own a dog,

Number that own a dog and a cat

$= 32 - 24$

$= 8$

(this can also be done similarly

with cats)

30. Geometry, NAP-32151

Racquet is 30 grid intervals long.

$30 \text{ intervals} = 60 \text{ cm}$

$\therefore 1 \text{ grid interval} = 2 \text{ cm}$

Racquet head is 13 grid intervals high,

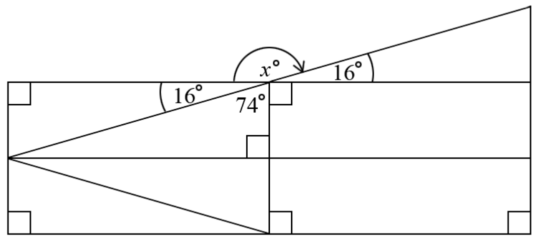
$$\begin{aligned}\therefore \text{ Head width} &= 13 \times 2 \\ &= 26 \text{ cm}\end{aligned}$$

31. Statistics and Probability, NAP-19200

Factors of 4 are: 1, 4, 2

$$\begin{aligned}\therefore P(\text{factor of } 4) &= \frac{3}{6} \\ &= \frac{1}{2}\end{aligned}$$

32. Geometry, NAP-61590



Using the right angle property of the rectangles,

$90^\circ - 74^\circ = 16^\circ$

Vertically opposite angles are equal,

$$\begin{aligned}\therefore x^\circ &= 180 - 16 \\ &= 164^\circ\end{aligned}$$